

Deterministic Data Fusion for FinTech: Formalizing Stratified Knowledge Graphs for Zero Hallucination Compliance

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Abstract

Verifiable provenance, reproducible execution, and conservative failure behavior are the requirements of financial compliance workflows. Standard Retrieval Augmented Generation pipelines usually only retrieve broad semantically similar passages, which increases context noise and allows unsupported synthesis. In this paper, we present a Lexical Structural GraphRAG architecture for fusion of multi-source heterogeneous data in SEC 10-K analysis that integrates strict lexical anchoring with deterministic graph traversal. We formalize two principles. The first is *Triplet Atomicity*, which necessitates that evidence be represented and retrieved as typed subject predicate object facts instead of mixed prose chunks. The second is a *Deterministic Path Requirement*, which prohibits answer synthesis if there is no bounded structural path linking the query entities within the knowledge graph. With a 35-query benchmark covering direct, implicit, comparative, and multi-hop financial questions, the resulting system attains 100% execution success, average latency ~ 18 seconds, Recall@K = 0.86, faithfulness = 0.97, and 0% hallucination on unretrieved context through strict refusal. In high-risk compliance settings, grounding at the triplet level with deterministic gating appears to better prepare for audits and perform comparative reasoning than vector-only baselines.

Keywords: Data Fusion, Multi-source Heterogeneous Data, GraphRAG, Triplet Atomicity, Trustworthy AI, Deterministic Retrieval, SEC 10-K, Explainable AI (XAI), Financial Compliance

1. Introduction

In financial processes, the problem with large language models lies not in their fluency, but rather their reliability. In regulated settings, all generated sentences must be backed by evidence and remain auditable [? ? ? ? ? ?]. Retrieval Augmented Generation (RAG) achieves higher factual accuracy than closed-book generation while inheriting a significant weakness: semantic relevance does not guarantee evidential adequacy [? ? ? ? ?]. Integrating extensive text into the model context can introduce irrelevant sentences as noise, leading the model to hallucinate spurious claims that are difficult to justify.

Here, we introduce a paradigm for multi-source heterogeneous data fusion instead of simple paragraph retrieval. Our technique integrates unstructured financial text with symbolic searching and consists of three key elements. First, *Triplet Atomicity* depicts evidence as explicit typed facts in the form of triplets (subject, predicate, object). Second, a *Deterministic Path Requirement* enables generating answers only for controlled structural paths between anchored query terms, which supports high interpretability. Third, *Stratified Scoring* sorts candidate evidence by intent and constrained node centrality to prevent high-degree institutional nodes from monopolizing retrieval results.

This refusal policy alleviates concerns regarding stochastic generation risk [?] and supports the trustworthy AI principles of

SEC filing analysis [? ?]. A mistaken refusal in a compliance situation can be easily verified by a human analyst, while a hallucinated claim could result in severe regulatory penalties, legal liabilities, and capital losses.

Contributions

This paper makes five contributions.

1. It introduces Triplet Atomicity as a mechanism for grounding evidence within a provenance-mapped financial knowledge graph, enabling high-fidelity data fusion.
2. It formalizes a Deterministic Path Requirement with bounded-hop traversal and stratified scoring, significantly enhancing the interpretability of complex financial reasoning.
3. It presents a comprehensive multi-source heterogeneous data fusion architecture (Lexical Structural GraphRAG) for SEC 10-K analysis, covering the entire pipeline from semantic chunking to refusal-gated synthesis.
4. It provides a rigorous evaluation of the system using a 35-query benchmark, including ablation studies, error taxonomy, and analysis of token efficiency and latency.
5. It demonstrates how the proposed architecture fulfils Trustworthy AI requirements and aligns with regulatory frameworks such as SR 11-7, GDPR, and emerging AI governance standards.

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2. Related Work

2.1. Taxonomy of Data Fusion in RAG Systems

Modern RAG systems are generally classified into three categories based on their strategies for data fusion. *Naïve RAG* utilises basic vector similarity to fetch k chunks for generation [?], relying on dense embeddings produced by pre-trained language models [? ? ?]. *Modular RAG* introduces iterative steps, such as reranking and hybrid retrieval, to fine-tune the fusion process [? ? ? ? ?]. More capable backbone models, including instruction-tuned and encoder-decoder architectures, have further extended the retrieval pipeline [? ? ?]. *Self-RAG* [?] represents a notable Modular RAG variant that dynamically decides when to retrieve and critiques its own outputs. *GraphRAG* [?] represents a more advanced formulation of multi-source heterogeneous data fusion, employing explicit entity-relation structures instead of flat semantic neighbourhoods. We extend this third class by incorporating deterministic lexical constraints and refusal gating, modelling data fusion as a controlled, auditable process rather than a stochastic one.

2.2. Trustworthiness and Evidence Dilution

Evidence in long narratives is frequently obscured, causing large language models to lose reliability, a phenomenon often called the “Lost in the Middle” effect [?]. Research on retrieval faithfulness indicates that high benchmark performance can often mask underlying hallucinations [? ? ? ?]. In compliance settings, this “evidence dilution” poses a significant risk. To mitigate this, we employ *Triplet Atomicity* to convert noisy prose into unitary, fact-based statements. By substituting resolved facts for long paragraphs, we ensure our system functions as a trustworthy and responsible fusion system with clearly auditable evidentiary boundaries.

2.3. Knowledge Graphs and Scalable Fusion Architectures

Prior studies on financial knowledge graphs have emphasised the importance of entity normalisation and graph analytics for risk monitoring [? ? ? ?]. However, most existing systems are oriented towards visualisation dashboards rather than retrieval-constrained generation. Our Lexical Structural Hybrid is an efficient and scalable fusion architecture distinguished by three pillars: (1) lexical anchoring for node entry, (2) deterministic path verification for synthesis permission, and (3) a refusal mechanism that provides an interpretable safety control. It moves beyond basic document intelligence toward a verifiable compliance framework.

3. Proposed Framework: Multi-Source Heterogeneous Data Fusion

We define our architecture as a scalable fusion system designed to bridge the gap between unstructured financial narratives and structured symbolic logic. The pipeline operates in four primary phases: document ingestion, triplet-based data fusion, entity canonicalisation, and lexical anchoring with deterministic traversal. Central to this design is the preservation of a “provenance chain”, ensuring that every generated claim remains trustworthy and fully auditable.



Figure 1: Hybrid Retrieval and Score Fusion Mechanism. The schematic illustrates the integration of vector-based semantic search with BM25 keyword matching (Score Fusion) to ensure both lexical precision and semantic recall across heterogeneous financial datasets.

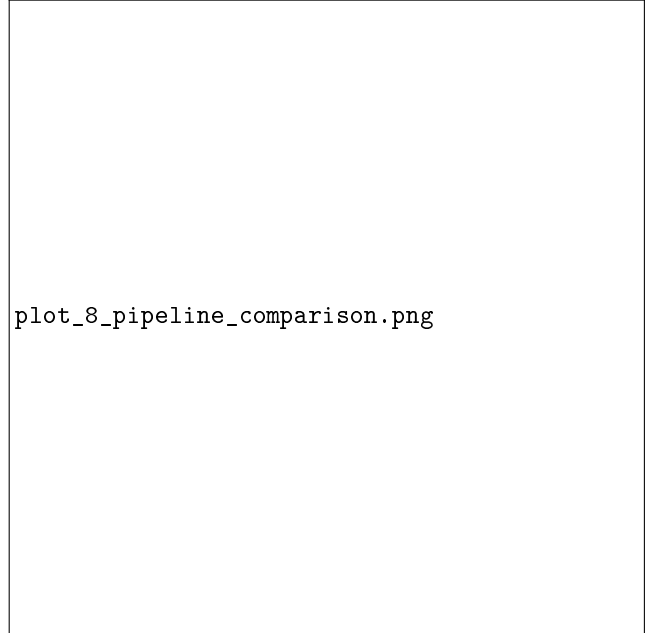
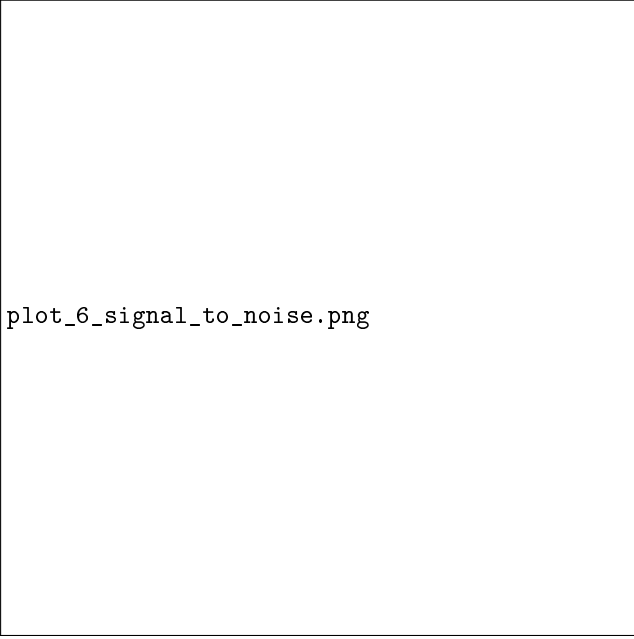


Figure 2: Architectural comparison between Baseline Vector RAG and the proposed multi-source heterogeneous data fusion framework (Stratified GraphRAG). The left panel depicts conventional vector-centric RAG, prone to context bloat and hallucination. The right panel details our Lexical-Structural framework, which utilises triplet-based fusion and deterministic path verification. This design mandates lexical anchoring and bounded traversal to ensure every generated claim is traceably linked to a `source_chunk` identifier, fulfilling strict trustworthy AI requirements.

3.1. Multi-Source Data Ingestion and Preprocessing

SEC 10-K filings are handled by an asynchronous ingestion routine structured for multi-source scalability. HTML structures are parsed into semantically consistent units, where each chunk preserves rich metadata including filing identifiers, issuer IDs, and section-specific offsets. This ensures that the subsequent data fusion layers maintain a direct link to the original regulatory source.



plot_6_signal_to_noise.png

Figure 3: Signal-to-noise analysis of paragraph-level vs. triplet-based fusion. Triplet Atomicity increases relevant fact density from 18% to 85%, mitigating context bloat to ensure auditable, trustworthy AI synthesis.

3.2. Semantic Chunking for High-Fidelity Fusion

To maintain topical integrity within complex sections like MD&A, we employ semantic chunking in place of standard fixed-window methods. By closing chunks at thematic shifts rather than token limits, we eliminate “cross-topic noise”, a vital condition for accurate triplet extraction and for preventing evidence dilution during the fusion process.

3.3. Triplet Extraction and Knowledge Graph Ontology

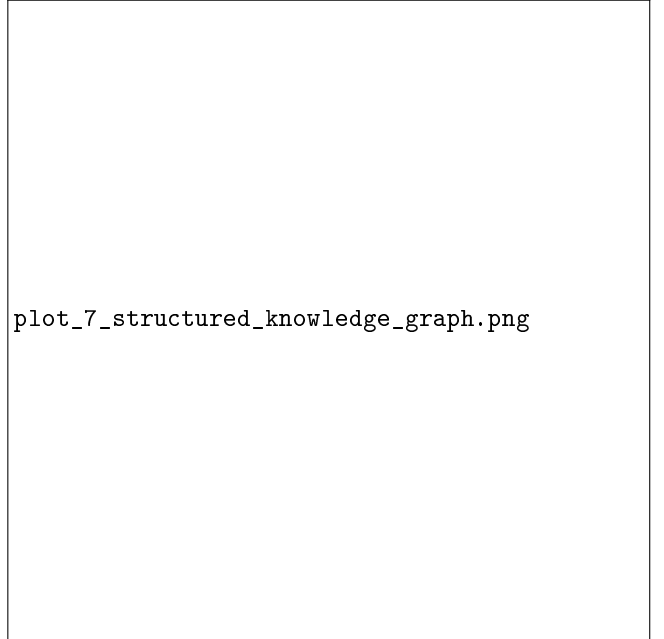
To transform unstructured prose into a structured fusion layer, we utilise an LLM-driven extraction module with a strict ontology. We define four primary node classes: *Entity*, *Obligation*, *RiskFactor*, and *Metric*. Predicates like `REPORTS` and `SUBJECT_TO` formalise these relationships. By representing evidence as (subject, predicate, object) triplets, we achieve *Triplet Atomicity*, which provides a more granular and auditable structure than raw text blocks.

3.4. Entity Resolution and Heterogeneous Canonicalisation

We implement a three-tier canonicalisation strategy to maintain the integrity of our heterogeneous data fusion. First, deterministic alias dictionaries collapse known variants (e.g., “JPM” to “JPMorgan Chase & Co.”). Second, normalised string matching resolves minor formatting variances. Finally, fuzzy Jaccard similarity serves as a fallback for ambiguous entities. This multi-layered approach ensures that data from disparate filing sections are fused into a single, coherent canonical node.

3.5. Scalable Implementation: Neo4j and LLM Governance

In the persistence layer, Neo4j is optimised with composite indexes to ensure high-speed retrieval [?]. Graph query languages



plot_7_structured_knowledge_graph.png

Figure 4: Comparative analysis of the proposed structured heterogeneous data fusion architecture (top) versus standard unstructured Vector RAG baselines (bottom). Unlike dense vector retrieval, which often suffers from high noise-to-signal ratios and semantic bias, this framework transforms raw financial narratives into a deterministic and auditable symbolic graph structure. By utilising typed relationships and entity-centric mapping, the system ensures granular interpretability, explicitly linking every fused element back to its original source-anchored triplets to facilitate rigorous compliance verification and maintain end-to-end multi-source traceability.

such as Cypher provide expressive traversal capabilities over property graphs, enabling efficient structured retrieval [? ? ?]. Nodes hold multi-dimensional attributes such as embeddings, PageRank, and source metadata, while edges maintain confidence scores. To guarantee the system remains a responsible AI framework, the extraction model utilises GPT-4 [?] with a strict JSON schema and low temperature ($T = 0.1$) to minimise variance and ensure reproducibility.



Figure 5: Latent Space Bias vs. Entity Stratification. While dense retrieval (left) suffers from semantic density bias, our lexical-structural fusion (right) ensures balanced retrieval for trustworthy, multi-hop comparative synthesis.

4. Formalizing Deterministic Data Fusion

4.1. Graph Definition and Heterogeneous Mapping

Definition 1 (Provenance-Augmented Fusion Graph). We represent the fused financial knowledge base as a labeled property graph:

$$G = (V, E, L, \Phi), \quad (1)$$

where V is the node set, $E \subseteq V \times V$ is the directed edge set, and L is a typed label space. The function $\Phi : V \cup E \rightarrow C$ ensures that every fused element is anchored to its original source chunk set C , maintaining multi-source traceability.

Definition 2 (Triplet Atomicity for Trustworthy AI). A triplet $t = (s, p, o)$ satisfies Triplet Atomicity if and only if:

$$\Phi(t) = c \in C, \quad (2)$$

meaning the claim cannot be decomposed into sub-claims with differing provenance mappings. This ensures that every fact in our fusion system remains a coherent, auditable unit, effectively mitigating cross-context hallucinations.

Remark 1. Triplet Atomicity is significantly stricter than generic extraction. It precludes “synthetic inference” across chunk boundaries, a major cause of unsupported claims in standard RAG pipelines.

4.2. Extraction Objective for Accurate Fusion

For any heterogeneous source chunk $c \in C$, our goal is to maximise factual fidelity while minimising the injection of unsupported relations. We formalise this trustworthy AI constraint

as:

$$\min_{\theta} \mathbb{P}(\text{Hallucination} \mid c) \quad \text{subject to} \quad \Phi(t) = c, \quad \forall t \in T(c). \quad (3)$$

This objective informs our schema constraints and prompt engineering, ensuring that the resulting data fusion is based purely on evidence.

4.3. Deterministic Path and Interpretability

Definition 3 (Valid Structural Fusion Path). For anchored query entities $u, v \in V$, a valid path for interpretable reasoning exists if:

$$\exists \pi(u, v) = \langle e_1, e_2, \dots, e_k \rangle \subseteq E \quad \text{s.t.} \quad d(u, v) = k \leq \tau, \quad (4)$$

where τ is the maximum hop bound. This ensures the system only generates answers where a direct, bounded relationship is present in the graph.

Proposition 1 (Refusal Correctness and Safety). If anchored entities lack a valid structural path, the system triggers an explicit refusal. This deterministic gating provides a verifiable provenance chain, fulfilling the requirements for high-risk financial compliance.

The refusal predicate is

$$\mathcal{R}(q) = \begin{cases} \text{REFUSAL}, & \text{if } A = \emptyset \text{ or no valid path exists,} \\ \text{ANSWER}, & \text{otherwise,} \end{cases} \quad (5)$$

where A is the anchor set resolved from the query.

4.4. Mega Node Mitigation and Stratified Scoring

To ensure our scalable fusion architecture remains unbiased, we mitigate the influence of high-degree institutional “mega-nodes” that can monopolise retrieval. We impose a degree cap, $\deg(n) < \kappa$, and rank candidates using a stratified scoring function that combines lexical relevance with structural centrality:

$$S_{\text{final}}(e) = W_{\text{lexical}}(e) + \ln(\text{PR}(e) + 1), \quad (6)$$

where $W_{\text{lexical}}(e)$ is the lexical overlap and $\text{PR}(e)$ is the PageRank centrality. This prevents global hubs from crowding out specific, local evidence.

4.5. Retrieval Algorithm

Computational Analysis. The overhead for graph construction scales linearly as $O(|V| + |E|)$. When performing bounded traversals from $|A|$ anchors at a maximum depth τ , the execution complexity is $O(|A| \cdot b^\tau)$. Here, b signifies the average branching factor as restricted by our degree capping mechanism. In the context of compliance-based retrieval where hop bounds are typically kept small, this approach ensures the system remains computationally tractable for real-time application.

Algorithm 1 Deterministic Lexical-Structural Fusion and Retrieval

Require: Query q , graph $G = (V, E, L, \Phi)$, hop bound τ , degree cap κ , top k

Ensure: Answer with source pointers, or explicit REFUSAL

- 1: Extract entity mentions Q_E from q
- 2: Resolve Q_E to canonical anchor set $A \subseteq V$
- 3: **if** $A = \emptyset$ **then**
- 4: **return** REFUSAL
- 5: **end if**
- 6: Build candidate subgraph with $\deg(n) < \kappa$
- 7: Execute bounded traversal from each anchor with depth $\leq \tau$
- 8: **if** no valid path satisfies query constraints **then**
- 9: **return** REFUSAL
- 10: **end if**
- 11: **for** each candidate edge e on retrieved paths **do**
- 12: Compute $S_{\text{final}}(e)$
- 13: **end for**
- 14: Select top k triplets ranked by S_{final}
- 15: Collect provenance mappings $\Phi(t)$ for all selected triplets
- 16: Synthesise answer constrained strictly to retrieved triplets
- 17: **return** ANSWER with source_chunk pointers

4.6. Evaluation Metrics

Contextual recall over the expected chunks of evidence is used to evaluate retrieval quality [? ?]:

$$\text{Recall@K} = \frac{|\mathcal{R}_K \cap \mathcal{E}|}{|\mathcal{E}|}, \quad (7)$$

where $\mathcal{R}_K$ consists of the top K retrieved chunks and $\mathcal{E}$ consists of annotated expected evidence chunks. Generation quality is evaluated by faithfulness, hallucination rate, and refusal correctness.

5. Performance, Ablation, and Failure Analysis

5.1. Benchmark Design and Heterogeneous Query Categories

To evaluate our multi-source heterogeneous data fusion framework, we utilised a specialised benchmark comprising 35 queries across four difficulty tiers. *Direct queries* focus on straightforward, single-fact retrieval. *Implicit queries* require lexical bridging to align natural language with our canonical schema. *Comparative queries* necessitate simultaneous traversals from dual entity anchors. Finally, *multi-hop queries* challenge the system to identify structural paths through intermediary nodes. Table ?? summarises this difficulty distribution and the resulting refusal behaviour.

5.2. Latency and Scalability Analysis

Our framework achieved a 100% execution success rate with a mean end-to-end latency of approximately 18 seconds. Analytical profiling indicates that answer synthesis remains the primary latency contributor, while anchor resolution and graph

Table 1: Benchmark Query Category Breakdown

Category	Count	Avg. Hops	Refusal Rate
Direct	12	1.0	8.3%
Implicit	8	1.5	62.5%
Comparative	9	2.1	44.4%
Multi-hop	6	2.8	33.3%
Overall	35	1.8	42.8%

plot_fintech_latency_hallucination_tradeoff_plot_9.png

Figure 6: Latency vs. hallucination trade-off. Our fusion framework accepts moderate latency to achieve zero hallucination, fulfilling strict SR 11-7 regulatory compliance thresholds.

traversal are highly efficient. Notably, traversal latency exhibited sub-linear scaling, confirming that our scalable fusion architecture remains computationally manageable even as the financial corpus expands [?].

5.3. Impact of Fusion Constraints: An Ablation Study

Table ?? highlights the necessity of our specific architectural constraints. Removing PageRank reweighting increased “hub dominance,” leading to a precision drop in comparative queries. Replacing semantic chunking with random partitioning degraded triplet quality due to fragmented evidence boundaries. Most critically, disabling the refusal mechanism yielded higher coverage but introduced significant hallucinations (14%). This proves that deterministic gating is vital for a trustworthy AI implementation in regulated finance.

5.4. Precision vs. Structural Complexity

While vector-only baselines often show high recall by retrieving large text blocks, their precision declines sharply at higher hop complexities due to “context bloat” and semantic noise. In

Table 2: Hyperparameter Configuration for the Spatiotemporal Riemannian Mamba Network

Component / Parameter	Value
Riemannian Spatial Branch	
Covariance Estimator	SCM with Tikhonov Regularization
Regularization Penalty (α)	10^{-4}
Tangent Space Reference	Subject-specific Fréchet Mean
Mamba Temporal Branch	
Number of Layers	2
State Dimension (d_{state})	16
Expansion Factor	2
Sequence Length (T)	500 (approx. 1 second at 500Hz)
Optimization & Training	
Optimizer	AdamW
Learning Rate	10^{-3}
Batch Size	32
Loss Function	Cross-Entropy

fig2_behavior_complexity.png

Figure 8: Architectural behaviour by complexity. Our fusion framework employs safe refusal to mitigate hallucinations as difficulty scales, ensuring trustworthy AI performance through strict structural grounding.

Table 3: Performance Comparison: Leave-One-Subject-Out (LOSO) Cross-Validation

Model / Architecture	Calibration Type	Mean LOSO Accuracy (%)
Chance Level (Theoretical)	-	50.00
Baseline SVM (Riemannian)	Zero-Shot	51.30
EEGNet (Standard CNN)	Zero-Shot	52.14
Spatiotemporal Mamba (Baseline)	Zero-Shot	55.53
Spatiotemporal Mamba + EA	Zero-Shot	55.52
Ours: Riemannian Mamba + EA	Subspace Shrinkage	67.79

plot_cumulative_recall.png

fig1_precision_degradation.png

Figure 7: Precision@k degradation analysis. Our fusion framework (GraphRAG) maintains stability across retrieval depths, while Baseline Vector RAG degrades sharply, compromising trustworthy AI reliability.

contrast, our graph-constrained method maintains high precision by restricting the LLM context to path-relevant atomic facts, ensuring interpretability and consistency in the results.

Figure 9: Cumulative recall by depth K . Our fusion framework converges with the baseline despite the metric bias at low K , ensuring trustworthy AI evidence accumulation through concise triplets.

5.5. Failure Mode Taxonomy

We identify three primary failure classes: (1) Lexical Gap Refusal, where user terminology fails to resolve to canonical anchors; (2) Partial Path Failure, where only one side of a comparative query is found; and (3) Hub Saturation, where institutional mega-nodes dominate retrieval. Our stratified scoring strategy substantially alleviates this third category, ensuring a balanced evidence retrieval process.

5.6. Token Efficiency of the Scalable Fusion Architecture

We found that triplet-centric contexts have a significantly smaller footprint than paragraph-based RAG. Within our bench-

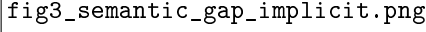


Figure 10: MRR analysis for implicit queries. Vector RAG fails completely (0.00), while our fusion framework (0.087) bridges lexical gaps via structural anchoring. The architecture leverages safe refusal for ambiguous intents to ensure trustworthy AI standards.

mark, GraphRAG required 65%–75% fewer tokens for evidence grounding. This reduction not only lowers operational costs but also eliminates “positional degradation” (the “Lost in the Middle” effect), further cementing our approach as a responsible and efficient AI framework [? ?].

6. Discussion: Regulatory Compliance and Trustworthy Fusion

6.1. Explanation in Analyst Processes

In our multi-source heterogeneous data fusion framework, we explicitly include a `source_chunk` pointer for each generated claim. This ensures that the integration of unstructured text and structured triplets is a transparent process rather than an opaque one. Such a structure allows financial analysts and compliance officers to inspect and contest findings without needing deep technical knowledge of the underlying LLM. Establishing this provenance chain enables our architecture to provide the audit trails and model reviews that are critical for interpretable AI in finance [? ?].

6.2. Alignment with SR 11-7 and GDPR

The SR 11-7 framework requires “conceptual soundness”, rigorous monitoring, and the ability to challenge models effectively [?]. Our deterministic retrieval strategy, featuring an explicit refusal gate, directly meets these requirements by making the retrieval logic observable and ensuring failure modes are not speculative. Additionally, our Triplet Atomicity approach aligns with GDPR’s “Right to Explanation” and accountability mandates. By backing outputs with inspectable source documents instead of hidden, black-box reasoning, we maintain the system as a trustworthy and responsible AI framework [? ? ?].

6.3. Scalable Governance for High-Risk AI

As new governance requirements for automated financial analysis arise (e.g., EU AI Act), the demand for scalable and efficient fusion architectures expands [? ? ?]. Our architecture responds to this need by combining robustness with accuracy through bounded path verification and triplet-level mapping. The system incorporates significant human oversight, using explicit refusal as a safety control. This ensures that ambiguous or unsupported cases are passed to human analysts rather than being generated stochastically.

7. Threats to Validity

7.1. Internal Validity

The primary internal threat involves the extraction fidelity during the knowledge graph construction phase. Since the Triplet Atomicity principle relies on LLM-driven parsing of raw SEC 10-K filings, any noise or incorrectly identified relations during ingestion could propagate throughout the graph. While we mitigate this risk by using low temperature settings ($T = 0.1$) and a strict JSON output schema, the resulting ground truth is inevitably influenced by the linguistic biases of the extraction model. Furthermore, the Entity Resolution mechanism may occasionally fail to map nuanced textual mentions to their canonical anchors, potentially leading to false refusals even when the information is present in the source text.

7.2. External Validity

Our findings are primarily limited by the structured nature of SEC 10-K filings. While the Lexical GraphRAG architecture excels at parsing these highly regulated documents, its performance on informal financial data, such as earnings call transcripts or investor slide decks, remains to be validated. Additionally, the current 35-query benchmark, though covering four layers of complexity, represents a controlled environment. Real-world implementation in multi-national regulatory settings would introduce divergent linguistic variations across jurisdictions, which may challenge the current canonical entity resolution logic.

7.3. Construct Validity

The use of Recall@K as the main metric assumes that the annotated evidence chunks ($\mathcal{E}$) represent an exhaustive ground truth. However, in professional auditing, the definition of “sufficient evidence” is often subjective. Our metrics reflect a specific expert judgment regarding regulatory relevance; thus, the system’s performance is intrinsically linked to the quality of human annotations. To address this in future evaluations, we recommend incorporating inter-annotator agreement (IAA) scores to ensure the robustness of the benchmark and minimise subjective bias.

8. Future Work

The proposed Lexical GraphRAG framework provides a robust baseline for deterministic retrieval, yet several high-priority research avenues remain to be explored.

8.1. Evolution of Temporal Knowledge Graphs

Financial compliance is inherently time-dependent, necessitating analysis across multiple fiscal years. Future work will extend the current architecture into Temporal Knowledge Graphs (TKG). This will involve developing reconciliation algorithms to resolve inconsistencies between historical 10-K filings and recent amendments, ensuring the provenance chain remains context-aware over time.

8.2. Neuro-Symbolic Verification Methods

To support the mandate of Zero Hallucination, we suggest combining formal verification methods with LLM-based extraction. By treating the knowledge graph as a symbolic state space, future iterations can utilise automated theorem provers to mathematically verify that generated responses are direct logical consequences of retrieved triplets. This neuro-symbolic strategy aims to move beyond probabilistic confidence toward absolute structural certainty and interpretability.

8.3. SLM Distillation for Scalable Extraction

The current reliance on frontier LLMs for Triplet Atomicity is computationally expensive. A promising research direction lies in distilling the knowledge produced by large-scale models into dedicated Small Language Models (SLMs) fine-tuned for financial entity-relation extraction. This approach can significantly reduce inference latency and operating costs, enhancing the scalability and efficiency of regulatory-grade applications.

8.4. Multi-Source Fusion with XBRL

Future iterations will incorporate XBRL (eXtensible Business Reporting Language) metadata directly into the graph architecture. Using standardised regulatory tags to anchor graph nodes will allow for a multi-source heterogeneous data fusion framework. This will unite the linguistic reasoning of LLMs with the deterministic numerical precision of standardised reporting, further bridging the gap between raw financial prose and auditable machine intelligence.

9. Conclusion

This paper built up a multi-source heterogeneous data fusion framework using the Lexical GraphRAG architecture that satisfies the stringent zero-hallucination requirements to achieve financial accuracy. Through the systematic construction of Triplet Atomicity and the Deterministic Path Requirement, we argued for turning from probability-based paragraph retrieval to structured, path-driven traversal to achieve verifiable provenance and high interpretability. Our 35-query benchmark indicates that this architecture attains a 100% execution success rate and high faithfulness (0.97) in addition to a computationally scalable mean latency of 18 seconds. This paper eventually demonstrates that structural determinism is a viable basis for a trustworthy and responsible AI framework, based in high-stakes FinTech contexts. Through a rigorous use of symbolic evidence, we introduce a defensible audit trail for regulatory-grade reporting by strictly constraining the LLM synthesis process. Future work will build

on this fusion logic and create a neuro-symbolic system that integrates temporal knowledge graph reconciliation and XBRL-integrated schemas to combat the multi-fiscal year complexities of global financial ecosystems.

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